

Specialist Maths (Bound Reference)

Aryan Gupta

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Index

	D	I
+	x^2	$\sin(3x)$
-	$2x$	$-\cos(3x)/3$
+	2	$-\sin(3x)/9$
-	0	$\cos(3x)/27$

This is a test page to play around with latex.
Continuing txt..

- Relation to the first term: $t_n = a + (n - 1)d$
- Relation to the previous term: $t_n = t_{n-1} + d$

1 Calculus

1.1 General Formulae:

Rule & Formula

Partial Fractions (Asymptotes)

Vertical Asymptote : *Denominator* = 0

Horizontal Asymptote : $\frac{\text{coefficient of numerator}}{\text{coefficient of denominator}}$

Power Rule

$$f'(x) = n * ax^{n-1}$$

Quotient Rule

$$f'(x) = \frac{u}{v}$$

$$f'(x) = \frac{u' * v - v' * u}{v^2}$$

Product Rule

$$f'(x) = u * v$$

$$f'(x) = u' * v + v' * u$$

Chain Rule

$$h(x) = f(g(x))$$

$$h'(x) = f'(g(x)) * g'(x)$$

$$\frac{dy}{dx} f(g(x)) = \frac{dy}{du} * \frac{du}{dx}$$

Examples

$$\text{Let } f(x) = \frac{2x + 3}{4 + 5x}$$

Find the asymptotes of f(x):

$$\text{Vertical Asymptote: } 4 + 5x = 0; x = -\frac{4}{5}$$

$$\text{Horizontal Asymptote: } \frac{2}{5}$$

$$f(x) = x^3 + x^2 + 5$$

$$f'(x) = (3)(x)^{3-1} + (2)(x)^{2-1} + (0)(5)$$

$$f'(x) = 3x^2 + 2$$

$$f(x) = \frac{x^3 + 2}{x}$$

$$f'(x) = \frac{(3x^2)(x) - (1)(x^3 + 2)}{x^2}$$

$$f'(x) = \frac{3x^3 - x^3 + 2}{x^2}$$

$$f'(x) = \frac{2x^3 + 2}{x^2}$$

$$f(x) = (x + 3)(x^4 + 5)$$

$$f'(x) = (1)(x^4 + 5) + (4x^3)(x + 3)$$

$$f'(x) = x^4 + 5 + 4x^4 + 12x^3$$

$$f'(x) = 5x^4 + 12x^3 + 5$$

$$f(x) = \sin(x^3 + 2x + 1)$$

$$f'(x) = \cos(x^3 + 2x + 1) * (3x^2 + 2)$$

$$f'(x) = (3x^2 + 2) \cos(x^3 + 2x + 1)$$

$$f(x) = \left(\frac{2x - 3}{4 + 5x} \right)^4$$

$$u = \frac{2x - 3}{4 + 5x}$$

$$\frac{dy}{dx} f(x) = \frac{dy}{du} * \frac{du}{dx}$$

$$\frac{dy}{dx} f(x) = 4 \left(\frac{2x - 3}{4 + 5x} \right)^3 * \left(\frac{2(4 + 5x) - 5(2x - 3)}{(4 + 5x)^2} \right)$$

$$\frac{dy}{dx} f(x) = 4 \left(\frac{2x - 3}{4 + 5x} \right)^3 \left(\frac{18}{(4 + 5x)^2} \right)$$

Integration (Power Rule)

$$\int ax^n dx$$

$$\int \frac{a}{n+1} x^{n+1} dx$$

Integration (Chain Rule OR U-Sub)

$$\int f(g(x))g'(x)dx$$

$$\int f(u)du$$

$$\begin{aligned} & \int x^3 + 5x + 2 dx \\ &= \frac{x^4}{4} + \frac{5x^2}{2} + 2x + C \end{aligned}$$

$$\begin{aligned} & \int \cos(x^2) 2x dx \\ & \int \cos(u) du \\ & \sin(u) + C = \sin(x^2) + C \\ & \int \frac{x}{x^2 + 1} \\ & \frac{1}{2} \int \frac{2x}{x^2 + 1} dx \\ & \frac{1}{2} \int \frac{1}{u} du \\ & \frac{1}{2} \ln(\|u\|) + C \\ &= \frac{1}{2} \ln(x^2 + 1) + C \end{aligned}$$

Integration (By Parts)

$$\int f(x)g'(x)dx$$

$$f(x)g(x) - \int g(x)f'(x)dx$$

$$\begin{aligned} & \int x \sin(x) dx \\ & u = x \\ & du = 1 \\ & v = -\cos(x) \\ & dv = \sin(x) \\ & -x \cos(x) - \int -\cos(x) dx \\ & -x \cos(x) + \int \cos(x) dx \\ & -x \cos(x) + \sin(x) + C \end{aligned}$$

Limits

$$\lim_{x \rightarrow a} \frac{1}{x} = \frac{1}{a}$$

$$\frac{a}{\infty} = 0$$

$$\begin{aligned} & \text{Let } f(x) = \frac{1-x}{x} \\ & \text{Find the value of } f(x) \text{ as } x \text{ approaches } 4. \\ & \frac{x \left(\frac{1}{x} - 1 \right)}{x} \\ & \frac{1}{\infty} = \frac{a}{\infty} = 0 \\ & 0 - 1 = -1 \\ & \therefore x = 0 \end{aligned}$$

Integration By Parts (D I Method)

Construct a table with columns D and I, what you're differentiating and integrating.

First Stop - 0 in D Column

Second Stop - Integrate Row

Third Stop - Row Repeats

Multiply Diagonally!

Stop 1:

$$\int x^2 \sin(3x) dx$$

$$= -\frac{1}{3}x^2 \cos(3x) + \frac{2}{9}x \sin(3x) + \frac{2}{27} \cos(3x) + C$$

	D	I
+	x^2	$\sin(3x)$
-	$2x$	$-\cos(3x)/3$
+	2	$-\sin(3x)/9$
-	0	$\cos(3x)/27$

Stop 2:

$$\int x^4 \ln(x) dx$$

$$= \frac{1}{5}x^5 \ln(x) - \int \frac{1}{5}x^4 dx$$

$$= \frac{1}{5}x^5 \ln(x) - \frac{1}{25}x^5 + C$$

	D	I
+	$\ln(x)$	x^4
-	$1/x$	$x^5/5$

Stop 3:

$$\int e^x \sin(x) dx$$

$$= -e^x \cos(x) + e^x \sin(x) - \int e^x \sin(x) dx$$

$$\int e^x \sin(x) dx + \int e^x \sin(x) dx =$$

$$-e^x \cos(x) + e^x \sin(x) - \int e^x \sin(x) dx + \int e^x \sin(x) dx$$

$$2 \int e^x \sin(x) dx = -e^x \cos(x) + e^x \sin(x)$$

$$= -\frac{1}{2}e^x \cos(x) + \frac{1}{2}e^x \sin(x) + C$$

	D	I
+	e^x	$\sin(x)$
-	e^x	$-\cos(x)$
+	e^x	$-\sin(x)$

Still an
Integral

1.2 Derivatives and Integrals of Trigonometric Functions

Derivatives

$$\sin(x) \longrightarrow \cos(x)$$

$$\cos(x) \longrightarrow -\sin(x)$$

$$\tan(x) \longrightarrow \sec^2(x)$$

$$\cot(x) \longrightarrow -\operatorname{cosec}^2(x)$$

$$\sec(x) \longrightarrow \sec(x) \cdot \tan(x)$$

$$\operatorname{cosec}(x) \longrightarrow -\operatorname{cosec}(x) \cdot \cot(x)$$

Integrals

$$\cos(x) \longrightarrow \sin(x) + C$$

$$\sin(x) \longrightarrow -\cos(x) + C$$

$$\sec^2(x) \longrightarrow \tan(x) + C$$

$$\sec(x) \cdot \tan(x) \longrightarrow \sec(x) + C$$

$$\operatorname{cosec}(x) \cdot \cot(x) \longrightarrow -\operatorname{cosec}(x) + C$$

$$\operatorname{cosec}^2(x) \longrightarrow -\cot(x) + C$$

1.3 Derivatives of Exponential Functions

Derivatives

$$\ln(u) \longrightarrow u'/u$$

$$\log_a(u) \longrightarrow u'/(u*\ln(a))$$

$$e^x \longrightarrow e^x$$

$$e^u \longrightarrow e^u*u'$$

$$a^x \longrightarrow a^x*\ln(a)$$

$$a^u \longrightarrow a^u*\ln(a)*u'$$

2 Series and Sequences

2.1 Types of Sequences and Series

Sequence/Series Type

Arithmetic Sequences

A sequence of terms with a common difference
(Addition)

Formulas:

Relation to the first term:

$$t_n = a + (n - 1)d$$

Relation to the previous term:

$$t_n = t_{n-1} + d$$

Where a is the first term and d is the common difference

Geometric Sequences

A sequence of terms with a common ratio
(Multiplication)

Formulas:

Relation to the first term:

$$t_n = a * r^{n-1}$$

Relation to the previous term:

$$t_n = r * t_{n-1}$$

Where a is the first term and r is the common ratio

General Formula of Recurrence

A formula applicable for both Geometric Sequences and Arithmetic Sequences.

Formula:

$$t_n = r * t_{n-1} + d$$

Arithmetic Sequence when $r = 1$

Geometric Sequence when $d = 0$

Arithmetic Series

The sum of the terms in an arithmetic sequence.

Formula:

Relation to the first term:

$$S_n = \frac{n}{2}(a + t_n)$$

Where a is the first term and n is the total number of numbers in the sequence.

Examples

Find the common difference of the sequence

5, 10, 15, 20, 25 ...

$$10 - 5 = 5$$

Common difference = 5

Find a general formula for this sequence.

$$t_n = 5 + 5(n - 1)$$

OR

$$t_n = t_{n-1} + 5$$

Hence, find the 50th term in this sequence.

$$t_{50} = 5 + 5(49) = 250$$

Find the arithmetic mean of the 49th and 50th term.

Find the common ratio of the sequence 6, 18, 54, 162 ...

$$\frac{54}{18} = 3$$

Common ratio = 3

Find a general formula for this sequence.

$$t_n = 6 * 3^{n-1}$$

OR

$$t_n = 3 * t_{n-1}$$

Hence, find the 15th term in this sequence.

$$t_{15} = 6 * 3^{14} = 2.869781 \times 10^7$$

Using the first principles, find a formula for the n th term of the sequence defined by the recurrence relation

$$t_n = t_{n-1} + 2n, t_1 = 5.$$

$$d = 2n$$

$$a = 5$$

$$t_n = 5 + 2n(n - 1)$$

Find the general formula for the sum of the sequence

3, 6, 9, 12, 15 ...

$$S_n = \frac{n}{2}(3 + t_n)$$

Find the sum of the first 60 terms in the sequence.

$$S_{60} = \frac{60}{2}(3 + t_{60})$$

$$S_{60} = 30(6 + 3(59))$$

$$S_{60} = 5400$$

Geometric Series

The sum of the terms in a geometric sequence.

Relation to the first term:

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

Where a is the first term r is the common ratio and n is the total number of terms in the sequence.

Xeno's Paradox (Geometric Series)

If $-1 < r < 1$, then the infinite geometric series $a + ar + ar^2 + \dots$ is convergent and the sum to infinity is given by:

$$S_\infty = \frac{a}{1 - r}$$

Where a is the first term and r is the common ratio.

Find the sum of the first 10 terms of the sequence

$$\left[\frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \dots \right]$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_{10} = \frac{\frac{1}{3} \left[\frac{1}{3}^{10} - 1 \right]}{\left(\frac{1}{3} - 1 \right)} = 1.5$$

Find the total time taken for a runner to travel from point A to B.

Assume that he is traveling at a constant speed and that he takes

1 minute to run half the distance from point A to B. Then he

takes $\frac{1}{2}$ minute to run half the remaining distance, and so on.

$$S_\infty = \frac{a}{1 - r}$$

$$S_\infty = \frac{1}{1 - \frac{1}{2}} = 2$$

3 Additonal Algebra

3.1 General Formulae

Formula

Linear Factor

$$\frac{ax+b}{(x+d)(x+e)} = \frac{A}{x+d} + \frac{B}{x+e}$$

Factorized Quadratic Factor

$$\frac{ax+b}{(x+c)(x+d)^2} = \frac{A}{x+c} + \frac{B}{x+d} + \frac{C}{(x+d)^2}$$

Quadratic Factor

$$\frac{ax^2+bx+c}{(x+d)(ex^2+fx+g)} = \frac{A}{x+d} + \frac{Bx+C}{ex^2+fx+g}$$

Example

$$\begin{aligned} & \frac{3x+5}{(x-1)(x+3)} \\ &= \frac{A}{x-1} + \frac{B}{x+3} \\ &= \frac{Ax+3A+Bx-B}{(x-1)(x+3)} \\ &= \frac{(A+B)x+3A-B}{(x-1)(x+3)} \\ &\Rightarrow A+B=3 \text{ \& } 3A-B=5 \\ &\therefore A=2 \text{ \& } B=1 \\ &= \frac{2}{x-1} + \frac{1}{x+3} \end{aligned}$$

$$\begin{aligned} & \frac{2x+10}{(x+1)(x-1)^2} \\ &= \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2} \\ &= \frac{A(x-1)^2+B(x+1)(x-1)+C(x+1)}{(x+1)(x-1)^2} \\ &= \frac{Ax^2-2Ax+A+Bx^2-B+Cx+C}{(x+1)(x-1)^2} \\ &= (A+B)x^2+(C-2A)x+A-B+C \\ &\Rightarrow A+B=0 \text{ \& } C-2A=2 \text{ \& } A-B+C=10 \\ &\therefore A=2, B=-2, C=6 \end{aligned}$$

$$\begin{aligned} & \frac{x^2+6x+5}{(x-1)(x^2+x+1)} \\ &= \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1} \\ &= \frac{Ax^2+Ax+A-Bx^2-Bx+Cx-C}{(x-1)(x^2+x+1)} \\ &= (A-B)x^2+(A-B+C)x+A-C \\ &\Rightarrow A+B=1 \text{ \& } A-B+C=6 \text{ \& } A-C=5 \\ &\therefore A=4, B=-3, C=-1 \end{aligned}$$

$$\begin{array}{cccccccc}
 & & & & 1 & & & \\
 & & & 1 & & 1 & & \\
 & & 1 & & 2 & & 1 & \\
 & 1 & & 3 & & 3 & & 1 \\
 & 1 & 4 & 6 & 4 & 1 & & \\
 1 & 5 & 10 & 10 & 5 & 1 & & \\
 1 & 6 & 15 & 20 & 15 & 6 & 1 & \\
 1 & 7 & 21 & 35 & 35 & 21 & 7 & 1
 \end{array}$$

$$\begin{aligned} & (x+3)^5 \\ & (1)(x^5)(3^0) + (5)(x^4)(3^1) + (10)(x^3)(3^2) \\ & + (10)(x^2)(3^3) + 5(x^1)(3^4) + (1)(x^0)(3^5) \\ & = x^5 + 15x^4 + 90x^3 + 270x^2 + 405x + 243 \end{aligned}$$

Example

1. Find the cost per person in terms of x .
2. If an extra 3 people go on the day trip, find the cost per person in terms of x .
3. It was calculated that, with the addition of three extra people, the cost of the trip per person was reduced by \$4. Find the number of people originally booked to go on the trip.

For the last 100km of the journey the salesman travelled at 10km/h faster than the previous 180km. The complete journey to Melbourne took 3 hours.

1. Write down an expression for the time taken for the first 180 km.
2. Write down an expression for the time taken for the last 100 km.
3. Hence determine the speeds of the salesman.

Solution

$$C_{\text{person}} = \frac{240}{x}$$
$$C_{\text{person}} = \frac{240}{x + 3}$$
$$C_{\text{person}} - 4 = \frac{240}{x + 3}$$

$$\frac{240}{x} = \frac{240}{x+3} + 4$$

$$x = 12, -15$$

$$\therefore x = 12$$

$$T_1 = \frac{180}{v_{10}}$$

$$T_2 = \frac{100}{v + 10}$$

$$\frac{180}{v} + \frac{100}{v+10} = 3$$

$$180(v + 10) + 100v = 3v(v + 10)$$

$$3v^2 - 250v - 1800 = 0$$

$v = 90$ for the first 180 km/h and 100 km/h for the last 100 km.

A tank is filled by two pipes. The smaller pipe alone will take 24 minutes longer than the larger pipe alone, and 32 minutes longer than when both pipes are used. How longer will each pipe take to fill the tank alone?
How long will it take for both pipes used together to fill the tank?

Flow:

$$S = \frac{\text{Capacity}}{\text{Time}} C$$

$$S_{\text{small}} = \frac{C}{x+24}$$

Let x = time for long pipe

$$S_{\text{large}} = \frac{C}{x}$$

$$S_{\text{Total}} = S_{\text{small}} + S_{\text{large}}$$

$$= \frac{C}{x+24} + \frac{C}{x}$$

$$= \frac{2Cx + 24C}{x(x+24)}$$

$$\text{Time} = \frac{\text{Capacity}}{\text{Flow}}$$

$$\frac{C}{2Cx + 24C}$$

$$\frac{x(x+24)}{x^2 + 24x}$$

$$t_{\text{total}} = \frac{x^2 + 24x}{2x + 24}$$

$$\text{Time:}$$

$$\text{Time} = \frac{\text{Capacity}}{\text{Flow}}$$

$$t_{\text{small}} + 24 = x + 24$$

$$t_{\text{total}} + 32 = \frac{x^2 + 24x}{2x + 24} + 32$$

$$x + 24 = \frac{x^2 + 24x}{2x + 24} + 32 = 24 \text{ mins}$$

$$\therefore t_{\text{large}} = 24$$

$$\therefore t_{\text{small}} = 48$$

$$t_{\text{total}} = \frac{24^2 + 24(24)}{2(24) + 24}$$

$$\therefore t_{\text{total}} = 16$$

4 Vectors

4.1 General Notes

- A directed line segment from a Point A to a point B is denoted by $\overrightarrow{AB}$
- The column $\begin{bmatrix} a \\ b \end{bmatrix}$ corresponds to the directed line segments which go 3 across and 2 up.
- Multiplication by a real number changes the length of a vector.
- The zero vector, denoted by 0, has no direction or magnitude.
- Collinear points: 3 or more points are collinear if they all lie on a single line.
- Concurrent lines: Three or more lines are concurrent if they all pass through a single point.

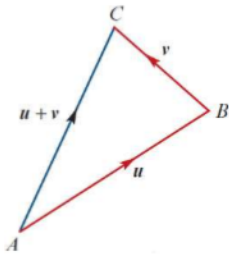
4.2 Geometry

Rule

Vector Addition

The sum of vectors u and v can be calculated by adding the two.

$$u + v = \overrightarrow{AC}$$



Example

What's $u + v$, if $u = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$

$$u + v = \begin{bmatrix} 4 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Vector Subtraction

To find $u - v$ we add $-v$ to u .

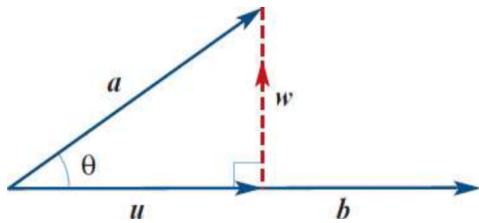
$$u - v = u + (-v)$$



For vectors $u = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ and $v = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$, find $2u - 3v$

$$\begin{aligned} 2 \begin{bmatrix} 3 \\ -1 \end{bmatrix} - 3 \begin{bmatrix} -2 \\ 2 \end{bmatrix} &= \begin{bmatrix} 6 \\ -2 \end{bmatrix} + \begin{bmatrix} 6 \\ -6 \end{bmatrix} \\ &= \begin{bmatrix} 12 \\ -8 \end{bmatrix} \end{aligned}$$

Perpendicular Vectors



$$a = u + w$$

$$u = kb$$

$$w = a - u = a - kb$$

Parallel Vectors

Two non zero vectors u and v are parallel if

there is some $k \in \mathbb{R} \setminus \{0\}$ such that

$$u = kv$$

If $u = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ and $v = \begin{bmatrix} -6 \\ 9 \end{bmatrix}$, then the vectors u and v are parallel as $v = 3u$.

4.3 Formulae

Formula

Unit Vector

$$|a|\hat{a} = a$$

$$\therefore \hat{a} = \frac{1}{|a|}a$$

Example

Let $a = 3i + 4j$

Find $|a|$, the magnitude of a , and hence find the unit vector in the direction of a .

$$1. |a| = \sqrt{3^2 + 4^2} = 5$$

$$2. \hat{a} = \frac{1}{5} * (3i + 4j)$$

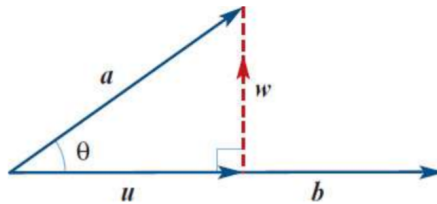
Perpendicular Vector

$$w \cdot b = 0$$

$$(a - kb) \cdot b = 0$$

$$k = \frac{a \cdot b}{b \cdot b}$$

Resolve $a = i + 3j$ into rectangular components, one of which is parallel to



$$b = 2i - 2j$$

$$a = u + w$$

$$u = u + (a - u)$$

$$= \frac{a \cdot b}{b \cdot b}b + \left[a - \frac{a \cdot b}{b \cdot b}b \right]$$

$$a \cdot b = -4$$

$$b \cdot b = 8$$

Parallel:

$$u = \frac{a \cdot b}{b \cdot b}b = -i + j$$

Perpendicular:

$$a - (-i + j) = 2i + 2j$$

$$\therefore i + 3j$$

Product of 2 Vectors

$$a \cdot b = a_1 b_1 + a_2 b_2$$

$$a \cdot b = |a||b| \cos(\theta)$$

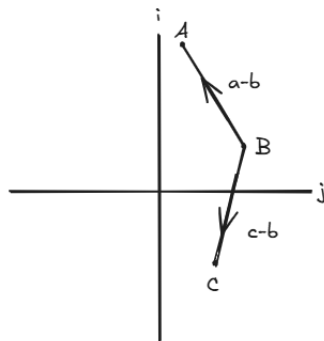
$$i \cdot i = 1, j \cdot j = 1$$

$$i \cdot j = 0$$

A, B and C are points defined by the position vectors, a, b and c respectively, where

$$a = i + 3j, b = 2i + j, c = i - 2j$$

Find the magnitude of $\angle ABC$.



$\angle ABC$ is the angle between $\overrightarrow{BA}$ and $\overrightarrow{BC}$

$$\cos \theta = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BA}| |\overrightarrow{BC}|}$$

$$\overrightarrow{BA} = a - b = i + 3j - (2i + j) = -i + 2j$$

$$\overrightarrow{BC} = c - b = i - 2j - (2i + j) = -i - 3j$$

$$\overrightarrow{BA} \cdot \overrightarrow{BC} = -5$$

$$|\overrightarrow{BA}| = \sqrt{(-1)^2 + (2)^2} = \sqrt{5}$$

$$|\overrightarrow{BC}| = \sqrt{(-1)^2 + (-3)^2} = \sqrt{10}$$

$$\Rightarrow \cos \theta = \frac{-5}{\sqrt{50}} = -\frac{1}{\sqrt{2}}$$

$$\therefore \angle ABC = \frac{3\pi}{4}$$

Vector Resolute

$$u = \frac{a \cdot b}{b \cdot b} b$$

Find the vector resolute of $a = i + 3j$ in the direction of $b = i - j$

$$u = \frac{(i + 3j)(i - j)}{|i - j|^2} (i - j) = \frac{-2}{2} (i - j) = -i + j$$

Scalar Resolute

$$a \cdot \hat{b} = a \frac{b}{|b|}$$

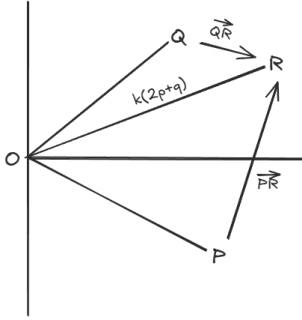
Find the scalar resolute or $a = 2i + 2j$ in the direction of $b = -i + 3j$

$$\frac{(2i + 2j)(-i + 3j)}{|-i + 3j|} = \frac{4}{\sqrt{10}}$$

4.4 Example

Question

Three points P, Q and R have position vectors p , q , $k(2p+q)$ respectively, relative to a fixed origin O.



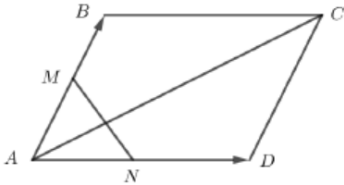
The points O, P and Q are not collinear. Find the value of k if:

- $\overrightarrow{QR}$ is parallel to p
- $\overrightarrow{PR}$ is parallel to q .
- P, Q and R are collinear.

Answer

$$\begin{aligned}
 \text{a. } \overrightarrow{QR} &= \overrightarrow{QO} + \overrightarrow{OR} \\
 &= -q + k(2p + q) \\
 &= 2kp - q(1 - k) \\
 \text{If } \overrightarrow{QR} \text{ is parallel to } \overrightarrow{OP}, & \\
 \text{then } \lambda \in R \setminus \{0\} \text{ such that} & \\
 2kp - q(1 - k) &= \lambda p \\
 \implies 2k &= \lambda, -q(1 - k) = 0 \\
 \therefore k &= 1, \lambda = 2 \\
 \text{b. } \overrightarrow{PR} &= \overrightarrow{PO} + \overrightarrow{OR} \\
 &= -p + k(2p + q) \\
 &= p(2k - 1) + kq \\
 p(2k - 1) + kq &= \lambda q \\
 \implies k &= \lambda, p(2k - 1) = 0 \\
 \therefore k &= \frac{1}{2}, \lambda = \frac{1}{2} \\
 \text{c. If P, Q, R are collinear,} & \\
 C \in R \setminus \{0\} \text{ such that:} & \\
 c\overrightarrow{PQ} &= \overrightarrow{QR} \\
 c(-p + q) &= 2kp - q(1 - k) \\
 c &= k - 1, c = -2k \\
 k &= \frac{1}{3}
 \end{aligned}$$

Suppose ABCD is a parallelogram. Let $\overrightarrow{AB} = u$ and $\overrightarrow{AD} = v$. Suppose M is the midpoint of AB and suppose N is the midpoint of AD.



- Let P be the point of intersection of lines AC and MN. By noting that $\overrightarrow{AP} = p\overrightarrow{AC}$ and $\overrightarrow{MP} = q\overrightarrow{MN}$ for some $p, q \in R$ show that $p = \frac{q}{2}$ and $\frac{q}{2} + p = \frac{1}{2}$. Hence, show that P divides $\overrightarrow{AC}$ in the ratio 1:3.

$$\begin{aligned}
 \text{a. } \overrightarrow{AP} &= \overrightarrow{AM} + \overrightarrow{MP} = p\overrightarrow{AC} \\
 &= \overrightarrow{AM} + q\overrightarrow{MN} = p(u + v) \\
 &= \frac{u}{2} + q\left(-\frac{u}{2} + \frac{v}{2}\right) = pu + pv \\
 &= \frac{u}{2} - q\frac{u}{2} + q\frac{v}{2} \\
 &= u\left(\frac{1}{2} - \frac{q}{2}\right) + \frac{q}{2}v = pu + pv \\
 p &= \frac{q}{2} \\
 P &= \frac{1}{2} - \frac{q}{2} \implies \frac{q}{2} + p = \frac{1}{2} \\
 \text{b. } 2p &= \frac{1}{2} \\
 \overrightarrow{AP} &= \frac{1}{4}\overrightarrow{AC} \\
 \overrightarrow{PC} &= \frac{3}{4}\overrightarrow{AC} \\
 \frac{\overrightarrow{AP}}{\overrightarrow{PC}} &= \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3} \\
 \therefore \text{Ratio} &= 1:3
 \end{aligned}$$

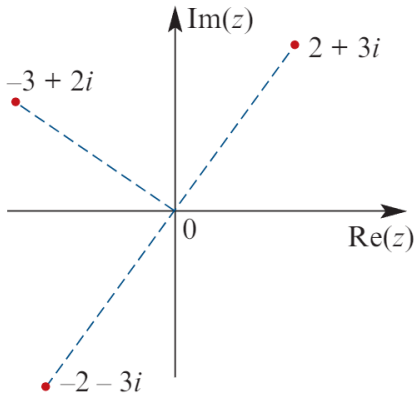
5 Complex Numbers

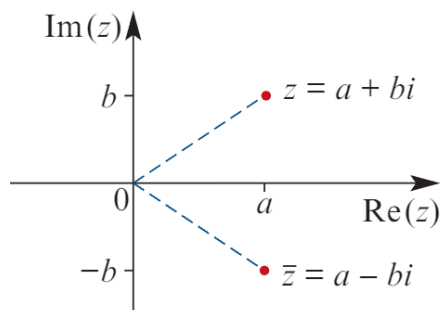
5.1 General Formulae

Formula	Example
Multiplication $z_1 = a + bi$ $z_2 = c + di$ $z_1 \cdot z_2 =$ $(ac - bd) + (ad + bc)i$	If $w = 3 - 2i$ and $z = 1 + i$ find wz $wz = (3 - 2i)(1 + i)$ $= 3 + 3i - 2i + 2$ $= 5 + i$
Conjugate $z = a + bi$ Conjugate of $z = \bar{z}$ $\bar{\bar{z}} = a - bi$ $\overline{w + z} = \bar{w} + \bar{z}$ $\overline{w \cdot z} = \bar{w} \cdot \bar{z}$	If $w = 2 - 3i$ and $z = -1 + 2i$ find $w + \bar{z}$ $= 2 - 3i + (-1 - 2i)$ $= 1 - 5i$
Modulus $z = a + bi$ $z\bar{z} = (a + bi)(a - bi)$ $= a^2 - abi + abi - b^2i^2$ $= a^2 + b^2$ $ z = \sqrt{a^2 + b^2}$ $z\bar{z} = z ^2$ $ wz = w z $	If $w = 2 + 3i$ and $z = 1 - 2i$ find $ wz $ $= w + z $ $= 13 + 5$ $= 18$
Division $\frac{z_1}{z_2} = z_1 z_2^{-1} = \frac{z_1 \bar{z}_2}{ z_2 ^2}$	Find $\frac{2 - i}{3 + 2i}$ $= \frac{(2 - i)(3 - 2i)}{(3 + 2i)(3 - 2i)}$ $= \frac{4 - 7i}{13}$

Rotation
Multiplication by -1 results in a rotation of 180° about the origin.
Multiplication by i results in a rotation of 90° anticlockwise about the origin.

$2 + 3i$ rotated 180° about the origin and 90° anticlockwise about the origin.



**Reflection**

The conjugate of a complex number

$z = a + bi$ is $\bar{z} = a - bi$.

Therefore $\bar{z}$ is the reflection of z in the horizontal axis.

Polar Form

$$z = x + yi$$

$$x = r \cos(\theta)$$

$$y = r \sin(\theta)$$

$$z = r \cos(\theta) +$$

$$(r \sin(\theta)i) =$$

$$r(\cos(\theta) + i \sin(\theta))$$

$$z = r \operatorname{cis}(\theta)$$

Express $z = 1 + i\sqrt{3}$ in polar form.

$$r = \sqrt{1 + 3} = 2$$

$$\theta = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$z = 2 \operatorname{cis}\left(\frac{\pi}{3}\right)$$

Calculating θ

$$\operatorname{atan2}(y, x) = \begin{cases} \arctan\left(\frac{y}{x}\right) & \text{if } x > 0 \\ \arctan\left(\frac{y}{x}\right) + \pi & \text{if } x < 0 \text{ and } y \geq 0 \\ \arctan\left(\frac{y}{x}\right) - \pi & \text{if } x < 0 \text{ and } y < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \text{ and } y > 0 \\ -\frac{\pi}{2} & \text{if } x = 0 \text{ and } y < 0 \\ \text{undefined} & \text{if } x = 0 \text{ and } y = 0 \end{cases}$$

OR

$$\theta = \begin{cases} \arccos\left(\frac{x}{r}\right) & \text{if } y \geq 0 \text{ and } r \neq 0 \\ -\arccos\left(\frac{x}{r}\right) & \text{if } y < 0 \\ \text{undefined} & \text{if } r = 0 \end{cases}$$

Multiplication in polar form

$$z_1 = r_1 \operatorname{cis}(\theta_1)$$

$$z_2 = r_2 \operatorname{cis}(\theta_2)$$

$$z_1 z_2 =$$

$$r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2)$$

Let $z_1 = 3 \operatorname{cis}\left(\frac{5\pi}{6}\right)$ and $z_2 = 2 \operatorname{cis}\left(\frac{5\pi}{6}\right)$. Find the product $z_1 z_2$ and express in Cartesian form.

$$z_1 z_2 = 6 \operatorname{cis}\left(\frac{5\pi}{3}\right)$$

Cartesian Form:

$$x = r \cos(\theta), y = r \sin(\theta)$$

$$x =, y =$$

Division in polar form

Inherit z_1 and z_2

from above.

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \operatorname{cis}(\theta_1 - \theta_2)$$

Let $z_1 = 3 \operatorname{cis}\left(\frac{5\pi}{6}\right)$ and $z_2 = 2 \operatorname{cis}\left(\frac{5\pi}{6}\right)$. Find the quotient of z_1 and z_2 .

$$\frac{z_1}{z_2} = \frac{3}{2} \operatorname{cis}(0)$$

6 Trigonometry

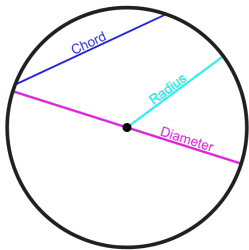
6.1 Key Terminology

Term

Defintions

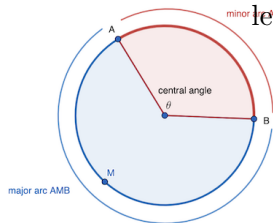
Chord

A chord of a circle is a straight line that connects any two points located on the circumference. If the chord passes through the center of the circle, then it is called a diameter.



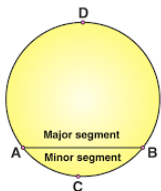
Arc

The arc of a circle is defined as the part or segment of the circumference of a circle. If the length of an arc is exactly half of the circle, it is known as a semicircular arc.



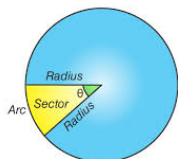
Segment

A segment of a circle is the region enclosed by a chord and the arc it subtends on the circle's circumference. It's like a "slice" of the circle's area formed by cutting the circle with a line (chord).



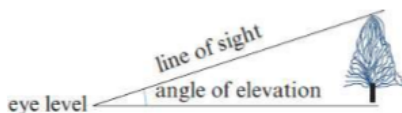
Sector

A sector of a circle is a portion of the circle enclosed by two radii and an arc.



Angle of Elevation

The angle of elevation is the angle between the horizontal and a direction above the horizontal.



Angle of Depression

The angle of depression is the angle between the horizontal and a direction below the horizontal.

Bearing

Direction measured clockwise from true North

Pythagorean Identity

$\sin^2 \theta + \cos^2 \theta = 1$

6.2 Exact Values

	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Undefined

6.3 Compound Angle Formulae

Compound Angle Formulae

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Double Angle Formulae

$$\sin(2A) = 2 \sin(A) \cos(A)$$

$$\cos(2A) = \cos^2(A) - \sin^2(A) = 1 - 2 \sin^2(A) = 2 \cos^2(A) - 1$$

$$\tan(2A) = \frac{2 \tan(A)}{1 - \tan^2(A)}$$

Half Angle Formulae

$$\sin\left(\frac{A}{2}\right) = \pm\sqrt{\frac{1 - \cos A}{2}}$$

$$\cos\left(\frac{A}{2}\right) = \pm\sqrt{\frac{1 + \cos A}{2}}$$

$$\tan\left(\frac{A}{2}\right) = \pm\sqrt{\frac{1 - \cos A}{1 + \cos A}} = \frac{\sin A}{1 + \cos A} = \frac{1 - \cos A}{\sin A}$$

6.4 Sums and Products

Products as Sums and Differences

$$2 \cos x \cos y = \cos(x - y) + \cos(x + y)$$

$$2 \sin x \sin y = \cos(x - y) - \cos(x + y)$$

$$2 \sin x \cos y = \sin(x + y) + \sin(x - y)$$

Sums and Differences as Products

$$\cos x + \cos y = 2 \cos\left(\frac{x + y}{2}\right) \cos\left(\frac{x - y}{2}\right)$$

$$\cos x - \cos y = -2 \sin\left(\frac{x + y}{2}\right) \sin\left(\frac{x - y}{2}\right)$$

$$\sin x + \sin y = 2 \sin\left(\frac{x + y}{2}\right) \cos\left(\frac{x - y}{2}\right)$$

$$\sin x - \sin y = 2 \sin\left(\frac{x - y}{2}\right) \cos\left(\frac{x + y}{2}\right)$$

6.5 General Formulae

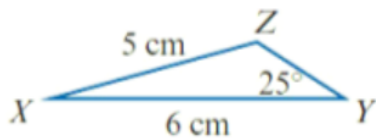
Formula

Sine Rule

$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$
$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$$

Example

Use the sine rule to find the magnitude of angle XYZ in the triangle, given that $Y = 25^\circ$, $y = 5$ cm and $z = 6$ cm.

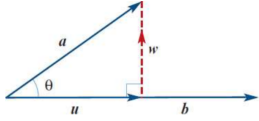


$$\frac{y}{\sin Y} = \frac{z}{\sin Z}$$
$$\Rightarrow Z = \sin^{-1}\left(\frac{Z \sin Y}{y}\right)$$
$$\sin^{-1}\left(\frac{6 \sin 25}{5}\right) = 30.47^\circ$$
$$\Rightarrow Z = 180 - 30.47^\circ = 149.53^\circ$$
$$\therefore Z = 30.47^\circ \text{ or } Z = 149.53^\circ$$

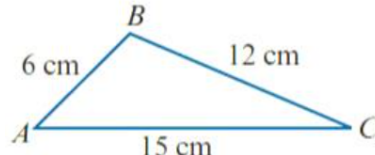
Cosine Rule

$$a^2 = b^2 + c^2 - (2bc * \cos(A))$$

$$\cos(A) = \frac{b^2 + c^2 - a^2}{2bc}$$



Find the magnitude of angle ABC.

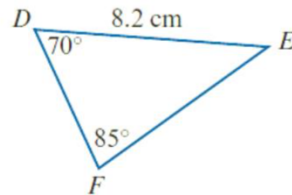


$$\begin{aligned} &= \cos^{-1} \left(\frac{a^2 + c^2 - b^2}{2ac} \right) \\ &= \cos^{-1} \left(\frac{12^2 + 6^2 - 15^2}{2 * 12 * 6} \right) \\ &= 108.21^\circ \end{aligned}$$

Area of a Triangle

$$\text{Area} = \frac{1}{2}bc \sin A$$

Find the area of the following triangle.



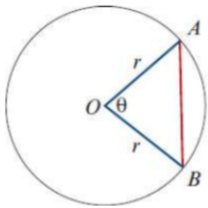
$$\begin{aligned} d &= \frac{8.2 \sin 70^\circ}{\sin 85^\circ} = 7.7349 \text{ cm} \\ A &= \frac{1}{2} 8.2 * 7.7349 \sin 25^\circ \\ A &= 13.403 \text{ cm}^2 \end{aligned}$$

Chord Length

In triangle OAP:

$$AP = r \sin \left(\frac{\theta}{2} \right)$$

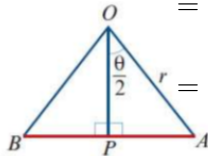
$$\therefore AB = 2r \sin \left(\frac{\theta}{2} \right)$$



The circle shown has center O and radius length 10 cm. The angle subtended at O by arc ACB has magnitude 120° . Find:

- the exact length of the chord AB

$$\begin{aligned} AB &= 2r \sin \left(\frac{\theta}{2} \right) \\ &= 2 * 10 \sin \frac{120}{2} = 20 \sin 60 \\ &= 20 * \frac{\sqrt{3}}{2} = 10\sqrt{3} \text{ cm} \end{aligned}$$



Arc Length

$$\text{Degrees: } \frac{\pi r \theta}{180}$$

$$\text{Radians: } r\theta$$

- the exact length of the arc ACB

$$\begin{aligned} l &= r\theta \\ &= 10 * \frac{2\pi}{3} = \frac{20\pi}{3} \text{ cm} \end{aligned}$$

- the magnitude of angle AOC, in degrees, if the minor arc AC has length 4 cm.

$$\begin{aligned} l &= r \frac{\pi \theta}{180} \\ 4 &= 10 \frac{\pi \theta}{180} \\ \Rightarrow \theta &= \frac{4 * 180}{10\pi} \\ &= 22.92^\circ \end{aligned}$$

Area of a sector

$$\text{Degrees: } \frac{\pi r^2 \theta}{360}$$

$$\text{Radians: } \frac{1}{2} r^2 \theta$$

4. the exact area of the minor sector AOB

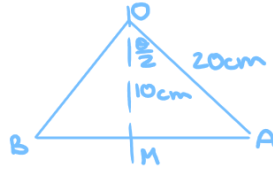
$$\begin{aligned} A &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} (10)^2 * \frac{2\pi}{3} = \frac{100\pi}{3} \text{ cm}^2 \end{aligned}$$

Area of a segment

$$\text{Degrees: } \frac{\pi r^2 \theta}{360} - \frac{1}{2} r^2 \sin \theta$$

$$\text{Radians: } \frac{1}{2} r^2 \theta - \frac{1}{2} r^2 \sin \theta$$

A circle, with center O and radius length 20 cm, has a chord AB that is 10 cm from the center of the circle. Calculate the area of the minor segment formed by this chord.



$$\begin{aligned} \cos \frac{\theta}{2} &= \frac{10}{20} = \frac{1}{2} \\ \frac{\theta}{2} &= \frac{\pi}{3} \\ \theta &= \frac{2\pi}{3} \\ A &= \frac{1}{2} r^2 \theta - \frac{1}{2} r^2 \sin \theta \\ &= \frac{1}{2} (20)^2 \left(\frac{2\pi}{3} - \sin \frac{2\pi}{3} \right) \\ &= \frac{400}{2} \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) \\ &= 200 \left(\frac{4\pi - 3\sqrt{3}}{6} \right) \\ &= \frac{400\pi - 300\sqrt{3}}{3} \\ &= \frac{100(4\pi - 3\sqrt{3})}{3} \end{aligned}$$

Unknown Lengths or Angles

$$a \cos x + b \sin x = r \cos(x - \alpha)$$

$$r = \sqrt{a^2 + b^2}$$

$$\cos \alpha = \frac{a}{r}$$

$$\sin \alpha = \frac{b}{r}$$

Express $\sqrt{3} \sin(2x) - \cos(2x)$ in the form $r \sin(2x + \alpha)$

$$a \cos x + b \sin x = r \sin(x + \beta), \quad r = \sqrt{a^2 + b^2}, \quad \cos \beta = \frac{b}{r} \quad \text{and} \quad \sin \beta = \frac{a}{r}$$

$$r = \sqrt{a^2 + b^2} = \sqrt{(\sqrt{3})^2 + (-1)^2} = \sqrt{4} = 2$$

$$\cos \beta = \frac{b}{r} = \frac{\sqrt{3}}{2}$$

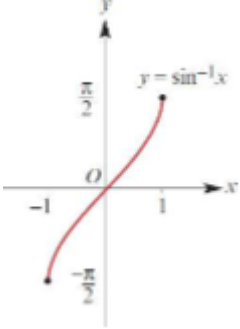
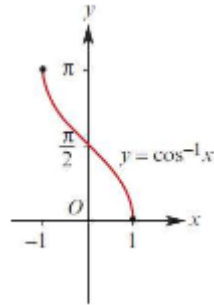
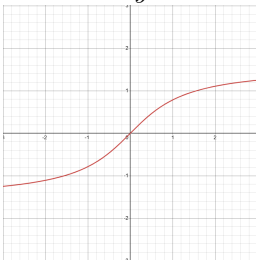
$$\sin \beta = \frac{a}{r} = -\frac{1}{2}$$

$$\therefore \beta = -\frac{\pi}{6}$$

$$f(x) = r \sin(2x + \beta) = 2 \sin \left(2x - \frac{\pi}{6} \right)$$

7 Graphing

7.1 Inverse Circular

Function	Domain	Range
$\sin^{-1} x = y$ 	$x \in [-1, 1]$	$y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x = y$ 	$x \in [-1, 1]$	$y \in [0, \pi]$
$\tan^{-1} x = y$ 	$x \in \mathbb{R}$	$y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Inverse relationships

- $\sin(\sin^{-1} x) = x$ where $x \in [-1, 1]$
- $\sin^{-1}(\sin x) = x$ where $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- $\cos(\cos^{-1} x) = x$ where $x \in [-1, 1]$
- $\cos^{-1}(\cos x) = x$ where $x \in [0, \pi]$
- $\tan(\tan^{-1} x) = x$ where $x \in \mathbb{R}$

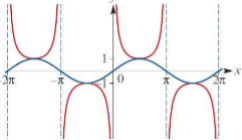
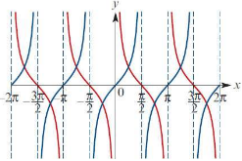
- $\tan^{-1}(\tan x) = x$ where $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

7.2 Reciprocal

- Local Maximum's become Local Minimum's and vice versa.
- If $y = f(x)$ is a non-zero polynomial function, then the graph of $y = \frac{1}{f(x)}$ will have vertical asymptotes where $f(x) = 0$
- If the graphs of a function and its reciprocal intersect, then it must be where $\%(x) = \pm 1$

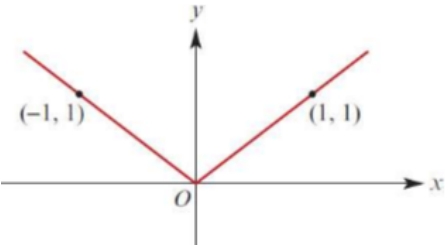
7.3 Reciprocal Circular

- Vertical asymptotes are intercepts of the original function.

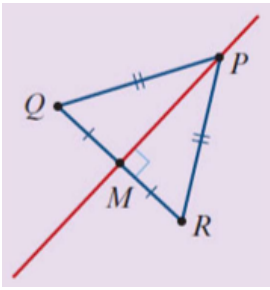
Function	Domain	Range
csc x and sec x 	$x \in \mathbb{R} \setminus \{\text{asymptotes}\}$	$(-\infty, \text{local minimum}] \cup [\text{local maximum}, \infty)$
cot x 	$x \in \mathbb{R} \setminus \{\text{asymptotes}\}$	$\mathbb{R}$

7.4 Modulus

- $|f(x)| = f(x)$ if $f(x) \geq 0$ and $|f(x)| = -f(x)$ if $f(x) \leq 0$



7.5 Locus of Points

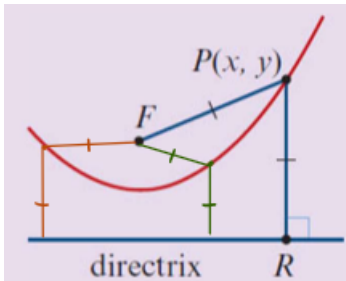


A circle is the locus of a point $P(x,y)$ that moves so that its distance (radius) from a fixed point $C(a,b)$ (centre) is constant.

- Radius is given by: $r = \sqrt{(x-a)^2 + (y-b)^2}$

An equidistant point is the same distance from 2 fixed points.

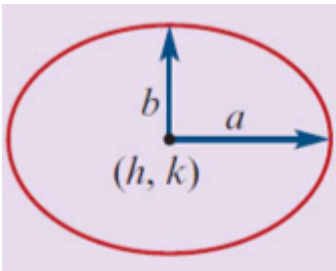
7.6 Parabolas



A parabola is the locus of a point P that moves so that its distance from a fixed point F is equal to its perpendicular distance from a fixed line.

- The fixed point is called the focus and the fixed line is called the directrix.

7.7 Ellipses



An ellipse is the locus of a point P that moves so that the sum of its distance from two fixed points F_1 and F_2 is a constant.

They have an equation of $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

- x-intercepts at $\pm a$
- y-intercepts at $\pm b$

7.8 Hyperbolas

A hyperbola is the locus of points of a point P that moves so that the difference between its distance from two fixed points F_1 and F_2 is a constant.

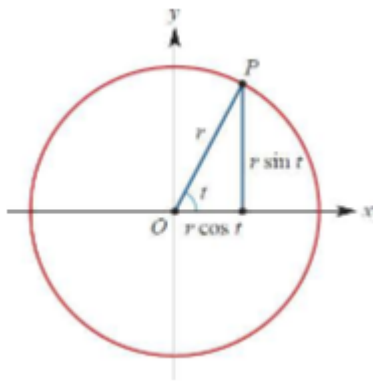
They have an equation of $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$

- Asymptotes at $y - k = \pm \frac{b}{a}(x - h)$

7.9 Parametric Equations

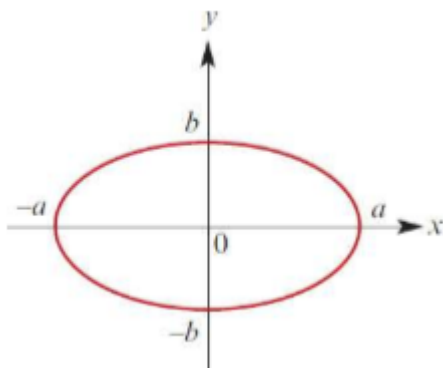
Solve for cartesian equation by eliminating the parameter, t .

Circles



- $x^2 + y^2 = r^2$
- $x = r \cos t$ and $y = r \sin t$
- $\cos^2 t + \sin^2 t = 1$

Ellipses

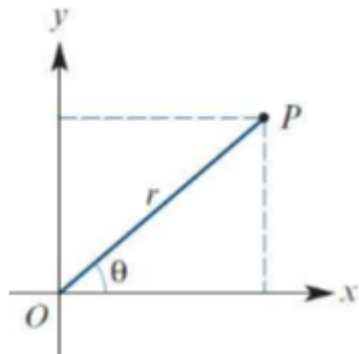


- $x = a \cos t$ and $y = b \sin t$
- $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \cos^2 t + \sin^2 t = 1$

Hyperbolas

- $x = a \sec t$ and $y = b \tan t$
- $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \sec^2 t - \tan^2 t = 1$

7.10 Polar Coordinates



If a point has polar coordinates $[r, \theta]$, then its cartesian coordinates are represented by;

- $x = r \cos \theta$ and $y = r \sin \theta$

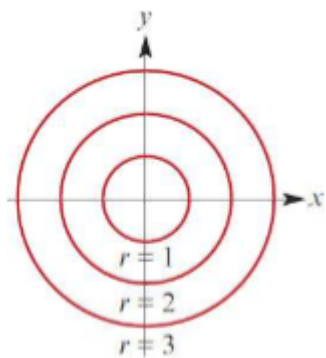
If a point has cartesian coordinates (x, y) then its polar coordinates are represented by;

- $r^2 = x^2 + y^2$ and $\tan \theta = \frac{y}{x}$

The point $P[r, \theta]$ can be described in infinitely many ways:

- $[r, \theta + 2n\pi]$ and $[-r, \theta + (2n + 1)\pi]$ for all $n \in \mathbb{Z}$

Circles



For a circle centered at the origin, its polar equation is $r = a$

Lines

For a straight line not passing through the origin the equation can be solved by substitution.

- $ax + by = c$
- $x = r \cos \theta$ and $y = r \sin \theta$

7.11 Examples

Question

17A

Find the implied domain and range of:

a. $y = \sin^{-1}(2x - 1)$

b. $y = 3 \cos^{-1}(2 - 2x)$

Answer

a. $\sin^{-1}(x), x \in [-1, 1]$

$$\sin(2x - 1) \implies -1 \leq 2x - 1 \leq 1$$

$$0 \leq x \leq 1$$

$$\therefore \text{Domain } [0, 1]$$

$$\text{Range } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

b. $3 \cos^{-1}(2 - 2x)$

$$-1 \leq 2 - 2x \leq 1 \implies \frac{3}{2} \geq x \geq \frac{1}{2}$$

$$\therefore \text{Domain } \left[\frac{1}{2}, \frac{3}{2}\right]$$

$$\text{Range } [0, 3\pi]$$

17B

Let $f(x) = 2 \cos x$ for $-2\pi \leq x \leq 2\pi$. Sketch the graphs of $y = f(x)$

and $y = \frac{1}{f(x)}$ on the same set of axes.

$2 \cos x$:

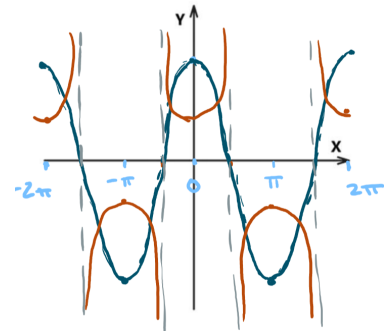
Dilated by a factor of 2 away from the x-axis.

$\frac{1}{f(x)}$ has vertical asymptotes when

$$2 \cos(x) = 0 \implies -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$$

$\frac{1}{f(x)}$, Local maximum when $2 \cos(x)$ at minimum (-2)

$$\left(-\pi, -\frac{1}{2}\right), \left(\pi, \frac{1}{2}\right) \implies \frac{1}{2 \cos(x)} = -\frac{1}{2}$$



17C

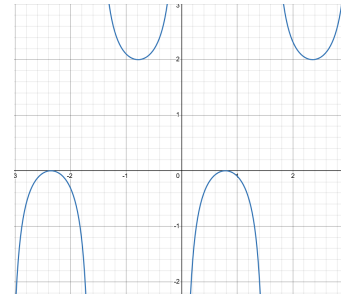
Describe a sequence of transformations that will take the graph of

$$y = \sec x \text{ to the graph of } y = -\sec\left(2x - \frac{\pi}{2}\right) + 1.$$

Sketch the transformed graph over the interval $[-\pi, \pi]$

Sequence:

1. Dilation by a factor of $\frac{1}{2}$ from the y axis.
2. Translation in the positive direction of the x-axis by $\frac{\pi}{4}$ units.
3. Reflection in the x-axis.
4. Translation in the positive direction of the y-axis by 1 unit.

**17D**

Solve the equation $|x^2 - 4x| = 3$ for x .

$$x^2 - 4x = \pm 3$$

$$x^2 - 4x = 3 \implies x = 2 \pm \sqrt{7}$$

$$x^2 - 4x = -3 \implies x = 1, x = 3$$

17E

Find the locus of a point $P(x, y)$ that moves so that it is equidistant from two points $Q(1, 1)$ and $R(3, 5)$.

Show that this is the perpendicular bisector of line segment QR .

$$\text{a. } QP = PR$$

$$\sqrt{(x-1)^2 + (y-1)^2} =$$

$$\sqrt{(x-3)^2 + (y-5)^2}$$

$$(x-1)^2 + (y-1)^2 = (x-3)^2 + (y-5)^2$$

$$x^2 - 2x + 1 + y^2 - 2y + 1 =$$

$$x^2 - 6x + 9 + y^2 - 10y + 25$$

$$8y = -4x + 32$$

$$y = -\frac{1}{2}x + 4$$

$$\text{b. Gradient of}$$

$$QR = \frac{\text{rise}}{\text{run}} = \frac{4}{2} \implies m_1 = 2$$

$$\text{Gradient of } P = -\frac{1}{2} \implies m_2 = -\frac{1}{2}$$

$$m_1 * m_2 = 2 * -\frac{1}{2} = -1$$

$\therefore$ Perpendicular

17F

Find the equation of the parabola with focus $F(0, c)$ and directrix $y = -c$.

Hence find the focus of the parabola with equation $y = 2x^2$

$$\text{a. } FP = PR$$

$$\sqrt{(x-0)^2 + (y-c)^2} =$$

$$\sqrt{(x-x)^2 + (y+c)^2}$$

$$x^2 + (y-c)^2 = (y+c)^2$$

$$x^2 + y^2 - 2yc + c^2 = y^2 + 2yc + c^2$$

$$-4yc = -x^2$$

$$y = \frac{1}{4c}x^2$$

$$\text{b. } y = \frac{1}{4c}x^2 = 2x^2$$

$$\frac{1}{4c} = 2$$

$$c = \frac{1}{8} \Rightarrow \text{Focus is } f\left(0, \frac{1}{8}\right)$$

17G

Find the equation of the locus of points $P(x, y)$ if the distance from P to the point $F(1, 0)$ is half the distance MP , the perpendicular distance from P to the line with equation $x = -2$

That is, $FP = \frac{1}{2}MP$

$$FP = \frac{1}{2}MP$$

$$\sqrt{(x-1)^2 + (y-0)^2} =$$

$$\frac{1}{2}\sqrt{(x+2)^2 + (y-y)^2}$$

$$(x-1)^2 + y^2 = \frac{1}{4}(x+2)^2$$

$$x^2 - 2x + 1 + y^2 = \frac{1}{4}(x^2 + 4x + 4)$$

$$4x^2 - 8x + 4 + 4y^2 = x^2 + 4x + 4$$

$$\Rightarrow 3x^2 - 12x + 4y^2 = 0$$

Complete the square:

$$\frac{(x-2)^2}{4} + \frac{y^2}{3} = 1$$

$\therefore$ center at $(2, 0)$

x-int at $x = 2$ and $x = 0$

y-int at $y = \sqrt{3}$ and $y = -\sqrt{3}$

17H

Consider the points $A(-2, 0)$ and $B(2, 0)$. Find the equation of the locus of points P satisfying $AP - BP = 2$

$$AP = 3 + BP$$

$$\sqrt{(x+2)^2 + (y-0)^2} =$$

$$3 + \sqrt{(x-2)^2 + (y-0)^2}$$

$$(x+2)^2 + y^2 =$$

$$9 + 6\sqrt{(x-2)^2 + y^2} + (x-2)^2 + y^2$$

$$8x - 9 = 6\sqrt{(x-2)^2 + y^2}$$

$$\text{For } x \geq \frac{9}{8}:$$

$$\Rightarrow 64x^2 - 144x + 81 = 36(x^2 - 4x + 4 + y^2)$$

$$28x^2 + 36y^2 = 63$$

$$\frac{28x^2}{63} - \frac{36y^2}{63} = 1$$

$$\frac{4x^2}{9} - \frac{4y^2}{7} = 1 \text{ for } x \geq \frac{3}{2}$$

Right side of a hyperbola defined by

$$\frac{4x^2}{9} - \frac{4y^2}{7} = 1.$$

17I

Find the points of intersection of the circle and line defined by the parametric equations:

Circle: $x = 5 \cos t$ and $y = 5 \sin t$

Line: $x = t - 3$ and $y = 2t - 8$

Circle:

$$\frac{x}{5} = \cos t \implies \frac{x^2}{25} = \cos^2 t$$

$$\frac{y}{5} = \sin t \implies \frac{y^2}{25} = \sin^2 t$$

$$\therefore \frac{x^2}{25} + \frac{y^2}{25} = 1$$

$$x^2 + y^2 = 25$$

Line:

$$y = 2(x + 3) - 8$$

$$\therefore y = 2x - 2$$

Substitution:

$$x^2 + (2x - 2)^2 = 25$$

$$5x^2 - 8x - 21 = 0$$

$$(x - 3)(5x + 7) = 0$$

Points of intersection:

$$(3, 4) \text{ and } \left(-\frac{7}{5}, -\frac{24}{5}\right)$$

17J

For the following pair of Cartesian coordinates, find two representations using polar coordinates, one with $r > 0$ and the other with $r < 0$.

$$(1, -\sqrt{3})$$

$$r = \sqrt{1 + (-\sqrt{3})^2} = 2$$

$$\tan \theta = \frac{-\sqrt{3}}{1} \implies \theta = -\frac{\pi}{3}$$

Polar Coordinate

$$\left[2, -\frac{\pi}{3}\right]$$

$$\left[-2, \frac{2\pi}{3}\right]$$

17K

A curve has polar equation $r = 2 \sin \theta$. Show that its cartesian equation is $x^2 + (y - 1)^2 = 1$

$$r = 2 \sin \theta$$

$$r^2 = 2r \sin \theta$$

$$y = r \sin \theta \implies r^2 = 2y$$

$$x^2 + y^2 = r^2$$

$$x^2 + y^2 = 2y$$

$$x^2 + y^2 - 2y + 1 - 1 = 0$$

$$\therefore r = 2 \sin \theta = x^2 + (y - 1)^2 = 1$$

8 Proofs

8.1 Calculus

Rule

Power Rule

$$f(x) = x^n$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$$

Proof

$$f(x) = x^2$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{h(2x + h)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} 2x + h = 2x$$

Product Rule

$$F(x) = f(x)g(x)$$

$$F(x+h) = f(x+h)g(x+h)$$

$$F'(x) = \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h}$$

$$F'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}$$

$$\text{Add and Subtract } f(x+h)g(x)$$

$$F'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x) + f(x+h)g(x) - f(x+h)g(x)}{h}$$

$$F'(x) = \lim_{h \rightarrow 0} f(x+h) \frac{g(x+h) - g(x)}{h} + g(x) \frac{f(x+h) - f(x)}{h}$$

By using the definition of a derivative:

$$F'(x) = f(x)g'(x) + g(x)f'(x)$$

Quotient Rule

8.2 Series and Sequences

Idea

Geometric Series

Proof

$$S_n = a + ar + ar^2 + ar^3 + \dots ar^{n-1}$$

$$rS_n = ar + ar^2 + ar^3 + \dots ar^{n-1} + ar^n$$

$$rS_n - S_n = S_n(r-1) = ar^n - a$$

$$S_n(r-1) = a(r^n - 1)$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

Zeno's Paradox

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{ar^n}{r - 1} - \frac{a}{r - 1}$$

$$S_n = \frac{ar^n}{r - 1} + \frac{a}{1 - r}$$

$$S_n = \lim_{n \rightarrow \infty} \frac{ar^n}{r - 1} + \frac{a}{1 - r}$$

Eliminating the first fraction:

$$\lim_{n \rightarrow \infty} \frac{ar^n}{r - 1} = \lim_{n \rightarrow \infty} \frac{a}{\frac{r - 1}{r^n}} = \frac{a}{\infty} = 0$$

$$S_n = \lim_{n \rightarrow \infty} \frac{a}{1 - r}$$

9 Cas Functions

Command & Syntax

$\text{cis}(\theta)$

Where θ is an angle.

$\text{cosec}(\theta)$

Where θ is an angle.

$\text{cot}(\theta)$

Where θ is an angle.

$\text{expand}(f, a)$

f is an expression in terms of x ,
and a is the variable used.

$\text{nmeth}(f, xa, n)$

f is an expression in terms of x
(execution of Program may
fail if there are other unknown
variables in the

expression)

xa is the initial guess for the
 x -value of a root of f

n is the number of iterations

$\text{scalares}(a, b)$

Where a, b are vectors

Returns the scalar resolute of a in
the direction of b .

$\text{sec}(\theta)$

Where θ is an angle.

Example

Find $\text{cis}(-\frac{2\pi}{7}) + \text{cis}(-\frac{12\pi}{7})$ in the form $A\cos(B\pi)$.

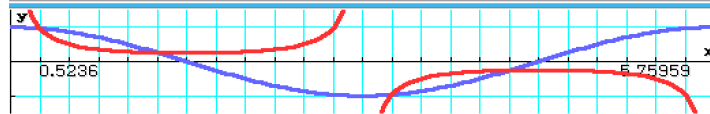
$$\text{cis}(\frac{2\pi}{7}) + \text{cis}(\frac{12\pi}{7})$$

$$2 \cdot \cos\left(\frac{2 \cdot \pi}{7}\right)$$

Find the largest set of values of x for which $\cos(x) > 14\text{csc}(x)$.

$$\text{solve}(\cos(x) = \frac{1}{4}\text{cosec}(x) \mid 0 < x < 2\pi, x)$$

$$\{x=0.26180, x=1.30900, x=3.40339, x=4.45059\}$$



Find the cotangent of $-\frac{2\pi}{3}$.

$$\text{cot}(-\frac{2\pi}{3})$$

$$\frac{\sqrt{3}}{3}$$

Split $\frac{3x+5}{(x-1)(x+3)}$ into 2 different fractions.

$$\text{expand}\left(\frac{3 \cdot x + 5}{(x-1) \cdot (x+3)}, x\right)$$

$$\frac{1}{x+3} + \frac{2}{x-1}$$

Find x_1, x_2 and x_3 using Newton's method with $x_0 = 0$ to approximate
an x -intercept of $h(x) = 2^x - x^2$.

$$\text{nmeth}(2^x - x^2, 0, 3)$$

$$\{-0.773, -0.013\}$$

Newton's method

$$\{i, x, y, dy/dx, \delta x\}$$

$$\{0.000, 0.000, 1.000, 0.693, -$$

$$\{1.000, -1.443, -1.713, 3.140$$

$$\{2.000, -0.897, -0.268, 2.166$$

$$\{3.000, -0.773, -0.013, 1.952$$

Find α for which the scalar resolute of $3\hat{i} + 2\hat{j} + \alpha\hat{k}$ in the direction
of $4\hat{i} - \hat{j} + \alpha^2\hat{k}$ equals $\frac{74}{\sqrt{273}}$.

$$\text{solve}(\text{scalares}([3, 2, \alpha], [4, -1, \alpha^2]) = \frac{74}{\sqrt{273}}, \alpha)$$

$$\{\alpha=4\}$$

Find the value of $\text{sec}(\theta)$ if $\tan(\theta) = \frac{5}{12}$.

$$\text{sec}(\tan^{-1}(\frac{5}{12}))$$

$$\frac{13}{12}$$

<i>suvat</i> (<i>a</i> , <i>b</i> , <i>c</i> , <i>d</i> , <i>e</i>)	Find v and t, if s=-80, u=2.5, a=-9.8
<i>a</i> , <i>b</i> , <i>c</i> are equations of the form g=h, e.g. s=4	<i>suvat</i> (s=-80 , u=2.5, a=-9.8, v, t)
<i>d</i> , <i>e</i> are the remaining variables from s , u , v , a , t	{t=-3.79, v=39.68}, {t=4.30, v=-39.68}
s is displacement, u is initial velocity, v is final velocity, a is acceleration, t is elapsed time	
<i>vectorrej</i> (<i>a</i> , <i>b</i>)	Find the vector rejection of $2\hat{i} - \hat{j} + 3\hat{k}$ perpendicular to $\hat{i} + \hat{j} - \hat{k}$.
Where a, b are vectors	<i>vectorrej</i> ([2, -1, 3], [1, 1, -1])
Returns the vector rejection of a perpendicular to b.	$\left[\frac{8}{3} \quad -\frac{1}{3} \quad \frac{7}{3} \right]$
<i>vectorres</i> (<i>a</i> , <i>b</i>)	Find the vector resolute of $3\hat{i} - 4\hat{j} + 12\hat{k}$ in the direction of $2\hat{i} + 2\hat{j} - \hat{k}$.
Where a, b are vectors.	<i>vectorres</i> ([3, -4, 12], [2, 2, -1])
Returns the vector resolute of a in the direction or b.	$\left[-\frac{28}{9} \quad -\frac{28}{9} \quad \frac{14}{9} \right]$

10 Miscellaneous Resources

Real Number System

